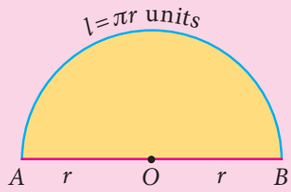
Note

1. The perimeter of a semi-circle

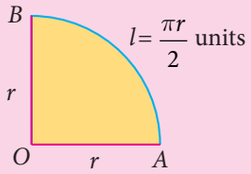
$P = l+2r$ *units*

$= \pi r + 2r = (\pi + 2)r$ *units*



2. The perimeter of a circular quadrant

$$P = l + 2r = \frac{\pi r}{2} + 2r = \left(\frac{\pi}{2} + 2\right) r \text{ units}$$



Example 2.1

The radius of a sector is 21*cm* and its central angle is 120°. Find (i) the length of the arc (ii) area of the sector (iii) perimeter of sector. $\left(\pi = \frac{22}{7}\right)$

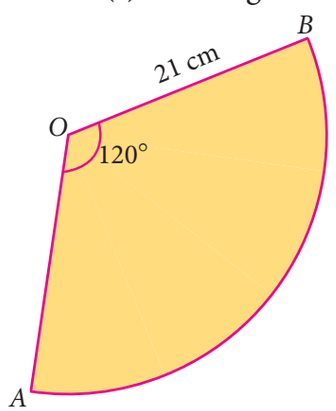


Fig. 2.13

Solution:

Radius, r = 21 *cm* and central angle, θ = 120°.

(i) Length of the arc, $l = \frac{\theta^\circ}{360^\circ} \times 2\pi r$ units

$= \frac{120^\circ}{360^\circ} \times 2 \times \frac{22}{7} \times 21$

= 44 cm (approximately)

(ii) Area of the sector, $A = \frac{\theta^\circ}{360^\circ} \times \pi r^2$ *sq.units.*

$= \frac{120^\circ}{360^\circ} \times \frac{22}{7} \times 21 \times 21$

A = 462 *sq.cm* (approximately)

Aliter:

Area of the sector

$A = \frac{lr}{2}$ *sq.units*

$= \frac{44 \times 21}{2} = 462\ cm^2$

(iii) Perimeter of the sector, $P = l + 2r$ *units*

= 44+2×21

= 44+42 = 86 *cm* (approximately)

Example 2.2

A circular shaped gymnasium ring of radius 35*cm* is divided into 5 equal arcs shaded with different colours. Find the length of each of the arcs.

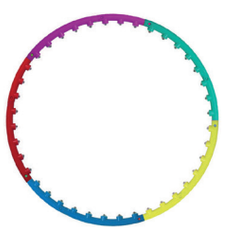


Fig. 2.14

Solution:

Radius, r = 35 *cm* and n = 5.

Length of each of the arcs, $l = \frac{1}{n} \times 2\pi r$ *units*

$= \frac{1}{5} \times 2 \times \pi \times 35 = 14\pi$ *cm.*

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